1 A student is investigating the maximum height reached by a light plastic ball when it is launched vertically from a compressed spring, as shown in Fig. 1.1.

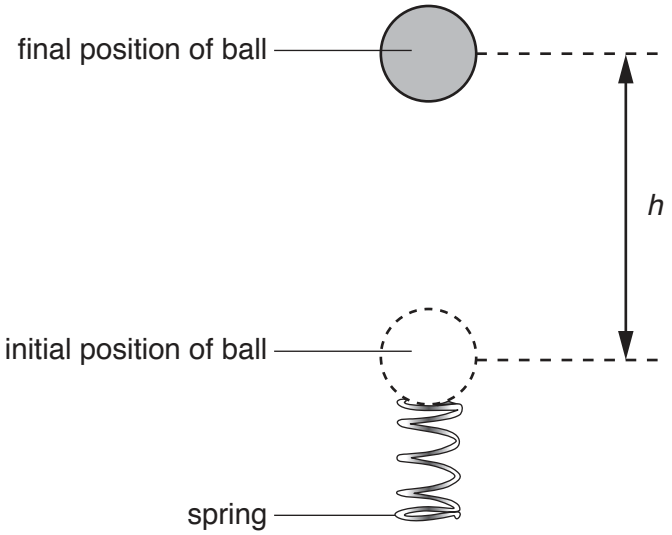


Fig. 1.1

It is suggested that the maximum height h of the ball and the compression x of the spring are related by the equation

$$\frac{4\pi r^3 \rho g h}{3} = \frac{1}{2} k x^2$$

where r is the radius of the ball, ρ is the density of the ball, g is the acceleration of free fall and k is the spring constant of the spring.

Design a laboratory experiment to test the relationship between h and x.
Explain how your results could be used to determine a value for ρ.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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